

CSCE 679 DATA VISUALIZATION · MODULE 10 · LECTURE NOTES

Deception

The deliberate failures, and the conventions that prevent them.

MODULE OUTCOMES

- Recognise the standard deceptive techniques.
- Explain the axis conventions and their reasons.
- Explain selective framing and aggregation.
- Distinguish deception from a design mistake.
- Review a figure for deception.

1 Axis manipulation

Technique	Its effect	The convention
Truncated bar axis	Exaggerates every difference in the figure.	Bars start at zero (Module 05 §4) — see the callout.
Inverted axis	Reverses the apparent direction of the trend.	Up means more, unless the inversion is labelled very loudly — and occasionally it is justified.
Dual axis	Any crossing or divergence can be manufactured by rescaling one side.	Use two stacked panels sharing an x axis (Module 05 §3).
Unlabelled log scale	Readers extrapolate linearly from a log plot.	Label it prominently, not only in the tick values.
Irregular tick spacing	Distorts the apparent rate of change.	Uniform spacing, or a clearly marked axis break.
Aspect ratio choice	Makes the same trend look flat or steep.	Bank to 45 degrees, or state the choice (Module 05 §4).

The axis techniques are checkable in seconds — look at the baseline, the direction, the scale type, and the tick spacing, which is four glances and catches most of this category. The four-glance check is the practical artefact of this section, and it is worth running on every figure you read rather than only on the suspicious ones, since the suspicious ones are the ones designed not to look it.

A TRUNCATED BAR AXIS IS A DECEPTION AND A TRUNCATED LINE AXIS IS NOT

A bar's length is read as a ratio from the baseline (Module 02 §1's shared-reference argument) — so **truncating the axis makes a 2% difference look like a 50% one, by an amount the reader cannot recover** even if they notice the axis label.

A line's position is read against the scale, which is labelled and gridded — so **a line chart with a non-zero axis is showing the variation at the resolution at which it actually occurs, which is frequently the correct choice** and is sometimes the only legible one.

Which means the rule has a reason rather than being an arbitrary convention — and **the reason tells you the exception: a line chart whose axis is truncated so far that the visible variation is measurement noise is also deceptive**, even though no bar was involved.

So the test is not 'does the axis start at zero' but 'does the encoding depend on the baseline' — **which is a question about the channel and settles both cases** and any new one, which is what a mechanism buys over a rule (CSCE 671 Module 05 §1).

2 Selective framing

- **Choosing the time window** — **any trend at all can be found in a long enough series by choosing the endpoints, and the resulting figure is entirely accurate**, which is what makes it effective.
- **Choosing the comparison** — against last month, last year, the same quarter two years ago, the forecast, or a competitor — **each of which gives a different story** from the same underlying number.
- **Choosing the aggregation level** — daily, weekly, or monthly; by region or by country — **which is Module 08 §3's modifiable areal unit problem applied to time and to categories**, and has the same consequence.
- **Omitting a series entirely**, which is **invisible: a reader cannot see what is not plotted**, and there is no trace in the figure that anything is missing.
- **And choosing the denominator** (Module 03 §3), **which can reverse a conclusion without any individual number being wrong. A reader cannot see what is not plotted — which makes omission the most effective and the least detectable technique in this module**, and the one no checklist of axis conventions catches.

WHICH MAKES SELECTIVE FRAMING HARDER TO DETECT THAN AXIS MANIPULATION

Every number in a selectively framed figure is correct, every axis is conventional, every scale is labelled, and **the deception is entirely in what was left out** of the frame.

So no amount of inspection of the figure itself detects it — you have to know the underlying data, or ask what the other windows and the other comparisons would show, which a reader generally cannot do and which is the asymmetry the technique exploits.

Which means the defence is procedural rather than visual: ask for the longer series, the alternative comparison, and the missing category — and treat an unusual window as a question rather than as a fact (why does this series begin in 2017?).

And in your own work, the defence is to show the alternatives — if three different windows tell the same story, show one and say that the others agree; and if they do not agree, that disagreement is the finding and belongs in the figure.

3 Deception and mistake

MOST BAD CHARTS ARE MISTAKES, AND THE TECHNIQUES ARE IDENTICAL EITHER WAY

A truncated axis produced by a plotting library's default is a mistake; the same truncated axis chosen deliberately to exaggerate a difference is a deception — and the resulting figure is identical in every respect.

So you cannot infer intent from a figure, and attempting to is both unfair and unnecessary: the correction is exactly the same either way, and the correction is what you wanted.

Which gives the useful position: name the technique and its effect, not the motive — 'this axis exaggerates the difference by a factor of ten' is checkable, and 'this chart is misleading' is an accusation that invites a defence rather than a fix.

And it is CSCE 671 Module 12's dual-use point in this subject — the same techniques serve honest simplification and deliberate deception — which is why the review in §4 looks for effects rather than for intentions and is therefore usable in a meeting.

4 Reviewing

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THE FOUR GLANCES (section 1)
  does the bar axis start at zero?
  which direction is "more"?
  is the scale linear or logarithmic, and is it
    labelled?
  are the ticks evenly spaced?

THEN THE FRAMING (section 2)
  what time window, and why that one?
  compared against what, and why that?
  what aggregation level?
  what is not shown?
  what is the denominator?

THEN THE UNCERTAINTY (Module 09)
  is there any, and is it shown?
  what is the sample size?

AND THEN THE ENCODING (Module 02 section 4)
  is the main variable on a high-ranked channel?

WHICH IS FOUR PASSES, AND TAKES A FEW MINUTES.
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Four passes, a few minutes — and it covers the deliberate deceptions, the honest omissions, and the encoding errors in one pass each, which is the full set of ways a figure can mislead. This checklist is the module's deliverable, and it composes the previous nine modules into something you can run on a figure you did not make, which is most of the figures you will encounter.

AND THE ASYMMETRY WORTH STATING

A figure is much easier to make misleading than to make honest, because the honest version requires stating the time window, the denominator, the spatial or categorical units, the uncertainty, and what is not shown — five things, each of which weakens the impression the figure makes.

So the incentive runs toward the misleading version wherever a figure is meant to persuade rather than to inform — which is **CSCE 671 Module 12 §4's measurement argument arriving in this subject**: the easy path moves the metric and the costs land elsewhere.

And the defence is the same as it was there: a figure that states its choices is checkable, and a reader who knows the checklist can ask the questions — which shifts the cost back to whoever framed it.

Which is why this module is a checklist rather than an ethics lecture — the technique names and the four passes are what actually transfer, and a general exhortation to be honest does not survive contact with a deadline.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Wilke, the pitfalls and the honesty chapters (free)	§1 and §2 , with the examples drawn and the axis arguments made from the channels.
Pandey et al. — How Deceptive are Deceptive Visualizations? (free)	§1 measured — how much a truncated axis actually changes a reader's judgement, which is a large amount.
Cairo — How Charts Lie	§2's selective framing in depth — and the best available treatment of the omission techniques. Library copy.
Huff — How to Lie with Statistics	§2 from 1954 , and still accurate — which is itself informative about how little the techniques have changed.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 Take one dataset and produce the most misleading honest figure you can**, using only framing.
- 2 Then produce the most honest one**, and list what you had to state.
- 3 Truncate a bar axis** and measure how much the apparent difference grows.
- 4 Ask five people to estimate the ratio** from each version.

- 5 **Find a published dual-axis chart** and rescale it to reverse the story.
- 6 **Find a trend by choosing endpoints** in a long series, three different ways.
- 7 **Find a figure where a series is omitted**, and say how you knew.
- 8 **Run the four glances** on ten published figures.
- 9 **Run all four passes** on one figure in detail.
- 10 **Rewrite one 'this chart is misleading' complaint** as a statement of effect.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Name six axis techniques and the convention for each.
- 2 Give the four glances.
- 3 Why is a truncated bar axis deceptive and a truncated line axis sometimes not?
- 4 State the test that settles both cases.
- 5 Name five selective framing techniques.
- 6 Why is omission the most effective, and what is the defence?
- 7 Why can you not detect selective framing from the figure?
- 8 Why can intent not be inferred, and what should you name instead?
- 9 Give the four review passes.
- 10 State the asymmetry and why the incentive runs the way it does.